Tutorial - 6 ✓ 10/3/26

(Ridge Regression)

Property features and prices

Bedrooms x_1	Age (years) x_2	price (₹ 100000s) (Y)
3	5	2
3	3	3

$2 = c + 3m_1 + 5m_2$ — (1)

$3 = c + 3m_1 + 3m_2$ — (2)

converting into matrix form

$$\underbrace{\begin{pmatrix} 2 \\ 3 \end{pmatrix}}_{Y} = \underbrace{\begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} c \\ m_1 \\ m_2 \end{pmatrix}}_{W}$$

$$W = (X^T X + \lambda I)^{-1} \cdot (X^T Y)$$

λ ↓ regularization parameter (take $\lambda = 1$)

$$X^T X \Rightarrow X = \begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix} \quad X^T = \begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix}$$

$$X^T X \Rightarrow \begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1+1 & 3+3 & 5+3 \\ 3+3 & 9+9 & 15+9 \\ 5+3 & 15+9 & 25+9 \end{pmatrix} = \begin{bmatrix} 2 & 6 & 8 \\ 6 & 18 & 24 \\ 8 & 24 & 34 \end{bmatrix}$$

$$X^TX + \lambda I = \begin{bmatrix} 3 & 6 & 8 \\ 6 & 18 & 24 \\ 8 & 24 & 34 \end{bmatrix} + 1 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{bmatrix}$$

$$(X^TX + \lambda I)^{-1} = \frac{\text{Adj}\,(X^TX + \lambda I)}{\det\,(X^TX + \lambda I)}$$

$$\text{Adj}\,(X^TX + \lambda I) \quad \begin{bmatrix} \begin{vmatrix} 19 & 24 \\ 24 & 35 \end{vmatrix} & \begin{vmatrix} 6 & 24 \\ 8 & 35 \end{vmatrix} & \begin{vmatrix} 6 & 19 \\ 8 & 24 \end{vmatrix} \\ \begin{vmatrix} 6 & 8 \\ 24 & 35 \end{vmatrix} & \begin{vmatrix} 3 & 8 \\ 8 & 35 \end{vmatrix} & \begin{vmatrix} 3 & 6 \\ 8 & 24 \end{vmatrix} \\ \begin{vmatrix} 6 & 8 \\ 19 & 24 \end{vmatrix} & \begin{vmatrix} 3 & 8 \\ 6 & 24 \end{vmatrix} & \begin{vmatrix} 3 & 6 \\ 6 & 19 \end{vmatrix} \end{bmatrix}$$

$$\Rightarrow \begin{vmatrix} 89 & 18 & 8 \\ 18 & 41 & 24 \\ 8 & 24 & 21 \end{vmatrix} = \begin{bmatrix} 89 & -18 & -8 \\ -18 & 41 & -24 \\ -8 & -24 & 21 \end{bmatrix}$$

$$\det\,(X^TX + \lambda I) = 3\,(19 \times 35 - 24 \times 24) - 6\,(6 \times 35 - 8 \times 24) + 8\,(6 \times 24 - 8 \times 19)$$

$$= 3\,(89) - 6\,(18) + 8\,(-8)$$

$$= 267 - 108 - 64 = 95$$

Now $(X^TX + \lambda I)^{-1} = \dfrac{\text{Adj}\,(X^TX + \lambda I)}{\det\,(X^TX + \lambda I)}$

$$\Rightarrow \frac{1}{95}\begin{bmatrix} 89 & -18 & -8 \\ -18 & 41 & -24 \\ -8 & -24 & 21 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{89}{95} & \frac{-18}{95} & -\frac{8}{95} \\ -\frac{8}{95} & \frac{41}{95} & \frac{24}{95} \\ -\frac{8}{95} & \frac{-24}{95} & \frac{21}{95} \end{bmatrix}$$

$$= \begin{bmatrix} 0.936 & -0.189 & -0.084 \\ -0.189 & 0.431 & -0.252 \\ -0.084 & -0.252 & 0.221 \end{bmatrix}$$

now $X^TY = \begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1\times2+1\times3 \\ 3\times2+3\times3 \\ 5\times2+3\times3 \end{pmatrix} = \begin{pmatrix} 5 \\ 15 \\ 19 \end{pmatrix}$

$$w = (X^TX + \lambda I)^{-1}(X^TY)$$

$$= \begin{bmatrix} 0.936 & -0.189 & -0.084 \\ -0.189 & 0.431 & -0.252 \\ -0.084 & -0.252 & 0.221 \end{bmatrix}\begin{bmatrix} 5 \\ 15 \\ 19 \end{bmatrix}$$

$$= \begin{bmatrix} 0.936\times5 + (-0.189\times15) + (-0.084\times19) \\ (-0.189\times5) + (0.431\times15) + (-0.252\times19) \\ (-0.084\times5) + (-0.252\times15) + (0.221\times19) \end{bmatrix}$$

$$\begin{bmatrix} 4.68 + (-2.835) + (-1.596) \\ (-0.945) + (6.465) + (-4.788) \\ (-0.42) + (-3.78) + (4.199) \end{bmatrix} = \begin{bmatrix} 0.249 \\ 0.732 \\ -0.001 \end{bmatrix}$$

c

m_1

m_2

$$w = \begin{pmatrix} 0.24 \\ 0.73 \\ -0.01 \end{pmatrix} \begin{matrix} - c \\ - m_1 \\ - m_2 \end{matrix}$$

$$\hat{y} = 0.24 + 0.73x_1 - 0.01x_2$$

$$\hat{y}_1 = 2.37$$

$$\hat{y}_2 = 2.38$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y-\hat{y}_i)^2$$

$$= \frac{1}{2}\left((2-2.37)^2 - (3-2.38)^2\right)$$

$$= 0.25 \approx 0.26$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y-\hat{y})^2}{\sum_{i=1}^{N}(y-\bar{y})^2} \quad \bar{y} = \frac{2+3}{2} = \frac{5}{2} = 2.5$$

$$= 1 - \frac{0.5213}{0.5} = -0.0426$$

$$RMSE = 0.50999$$

$\Rightarrow c = 0.254,\ m_1 = 0.747,\ m_2 = 0.018$

$$\hat{y}_1 pre = 0.254 + 3(0.747) + 5(0.018) = 2.585$$

$$\hat{y}_2 pre = 0.254 + 3(0.747) + 3(0.018) = 2.544$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 = \frac{1}{2}\left[(2-2.585)^2 + (3-2.544)^2\right]$$

$$= 0.2729$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \Rightarrow 1 - \frac{0.5457}{0.5} = -0.0914 \qquad \bar{y} = 2.5$$

$$RMSE = 0.522399$$